Basic Questions

Antibiogram

Full Board

Cases

ATYPICALS	MRSA	ENTEROCOCCUS	GPCs	GNRs	PSEUDOMONAS	ANAEROBES Oral	Gut	ESBLs

Antibiotics: Spectrum of Activity

Basic Questions

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Cases

ATYPICALS	MRSA	ENTEROCOCCUS	GPCs	GNRs	PSEUDOMONAS	ANAEROBES Oral	ANAEROBES Gut	ESBLs

Click each button until the arrow disappears.

1

Does it cover gram positives (*Staph*, *Strep*)?

Does it cover MRSA?

Does it cover Enterococcus?

2

Does it cover gram negatives (enteric GNRs)?

Does it cover *Pseudomonas*?

Does it cover *ESBL* organisms?

3

Does it cover atypicals (*Mycoplasma*, *Chlamydophila*, *Legionella*)?

4

Does it cover anaerobes?

Does it cover gut anaerobes *(B. fragilis)?*

Basic Questions

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ATYPICALS	MRSA	ENTEROCOCCUS	GPCs	GNRs	PSEUDOMONAS	ANAEROBES Oral	Gut	ESBLs

Click on each antibiotic class to learn more

Penicillins

Cephalosporins

Carbapenems

FQ

Anti-PsA

Macrolides

CA-MRSA

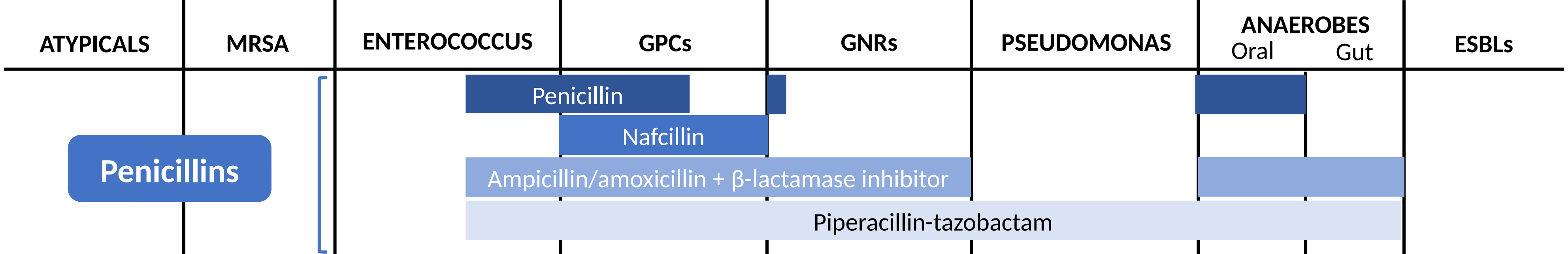
Hospital MRSA

Basic Questions

Antibiogram

Full Board

Cases



Cephalosporins

Carbapenems

FQ

Anti-PsA

Macrolides

CA-MRSA

Hospital MRSA

General Rules:

1. No MRSA
2. No atypical coverage
3. ↑ GNR coverage

Penicillin

1st line for *Strep*, syphilis
↑ *Staph* resistance

Nafcillin

Good for *Staph* and *Strep*
1st line for MSSA

Ampicillin

Broad spectrum
+ GN, gut anaerobes
β-lactamase inhibitors ↑ GN and *Staph* coverage

Piperacillin

+ PsA coverage

Basic Questions

Antibiogram

Full Board

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ATYPICALS | MRSA | ENTEROCOCCUS | GPCs | GNRs | PSEUDOMONAS | ANAEROBES (Oral, Gut) | ESBLs

Penicillins

Cephalosporins

1st G (cefazolin)

2nd G (cefotetan, cefuroxime)

3rd G (ceftriaxone, cefpodoxime)

3rd G (ceftazidime)

4th G cefepime

5th G ceftaroline

Carbapenems

FQ

Anti-PsA

Macrolides

CA-MRSA

Hospital MRSA

General Rules:

1. No *Enterococcus*
2. No atypicals
3. ↑ GNR coverage with later gen

1st G — Good for *Staph* and *Strep*

2nd G — + some GNRs (*Klebsiella, Moraxella, H. flu*)

3rd G — **Ceftriaxone, cefpodoxime:** + GN (*Enterobacteriaceae*)

Ceftazidime: poor GP (no *Staph*), + PsA

4th G — Broad GP and GN, including PsA

"5th G" — + MRSA coverage

Basic Questions

Antibiogram

Full Board

Cases

General Rules:

1. Broadest antibiotic class
2. Most reliable ESBL coverage
3. No MRSA, poor *Enterococcus*
4. No atypical coverage

Ertapenem

GP, GN, anaerobic coverage
No PsA
Once daily dosing

Meropenem

+ PsA coverage

ATYPICALS | MRSA | ENTEROCOCCUS | GPCs | GNRs | PSEUDOMONAS | ANAEROBES (Oral, Gut) | ESBLs

Penicillins

Cephalosporins

Carbapenems

Ertapenem

Ertapenem

Imipenem, meropenem, doripenem

FQ

Anti-PsA

Macrolides

CA-MRSA

Hospital MRSA

Basic Questions

Antibiogram

Full Board

Cases

General Rules:

1. Good GNR coverage
2. Atypical coverage
3. High bioavailability

Cipro

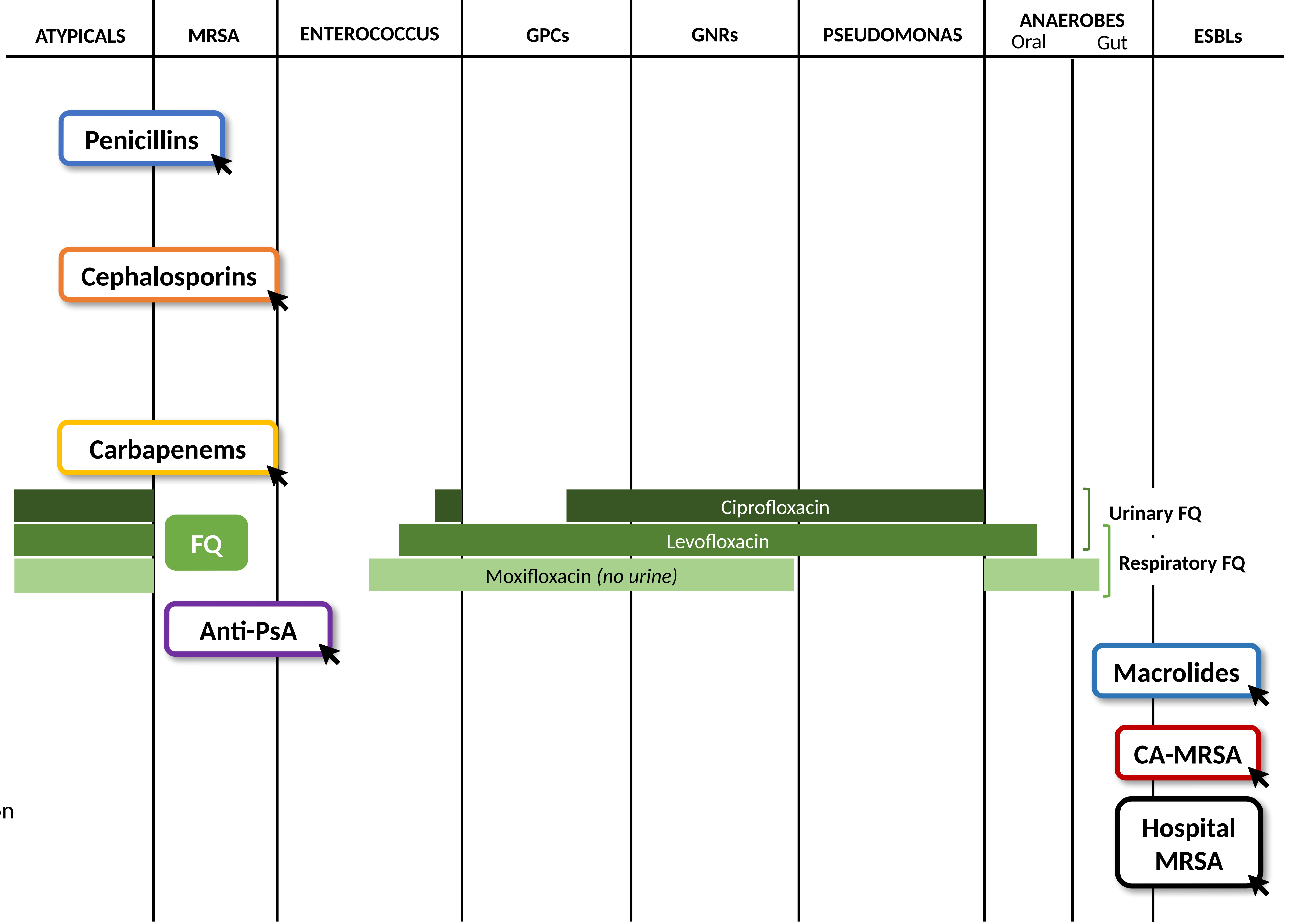
Urine
Minimal *Staph/ Strep* coverage
Can be used for atypical or PsA coverage in lung infections

Levo

Urinary + respiratory + *Staph and Strep* coverage

Moxi

Respiratory
Poor urine penetration
Broadest anaerobic coverage



Basic Questions

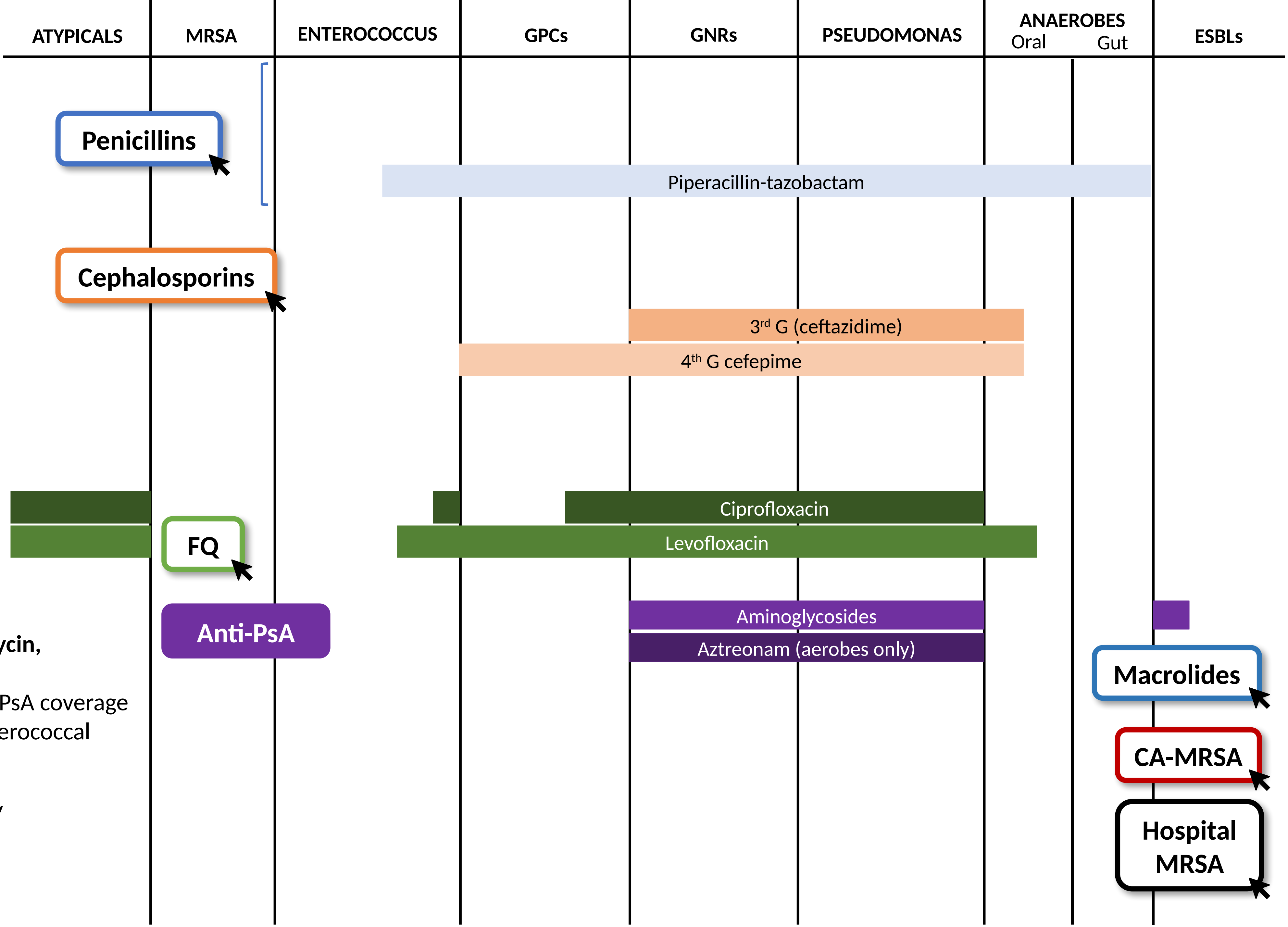
Antibiogram

Full Board

Cases

Piperacillin-tazobactam, later generation cephalosporims, and FQs are commonly used for anti-Pseudomonal coverage.

Additional anti-Pseudomonal agents include the following:



AGs

Gentamicin, tobramycin, plazomicin

Good aerobic GNR + PsA coverage

Some have ESBL, Enterococcal activity

Aztreonam

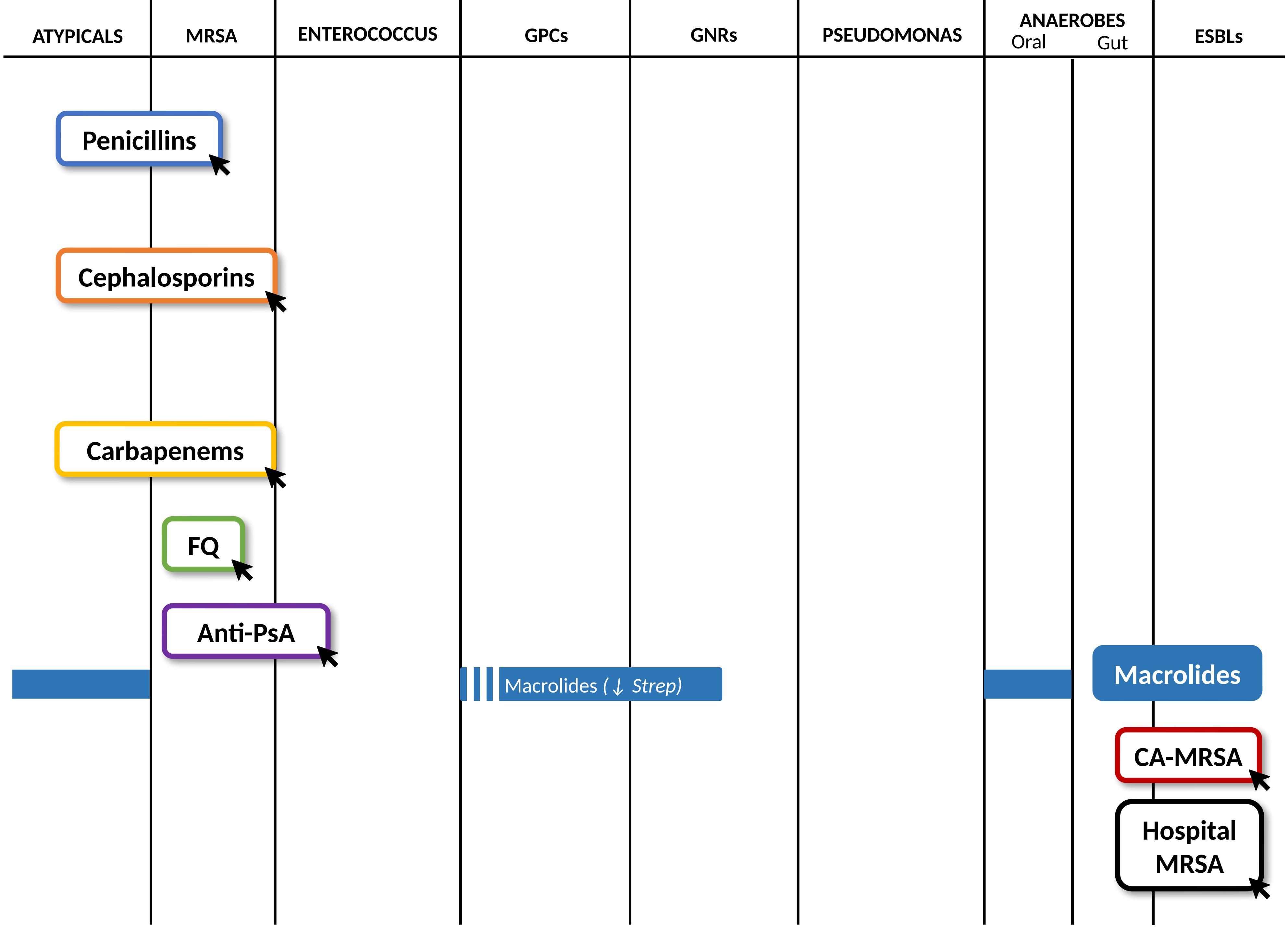
GN aerobes only + PsA coverage

Basic Questions

Antibiogram

Full Board

Cases



Azithromycin, Erythromycin, Clarithromycin

Good atypical coverage

Some GP coverage, though ↑ *Strep* resistance

Basic Questions
Antibiogram
Full Board
Cases

General Rules:
Coverage should be based on local antibiotic resistance

Clindamycin

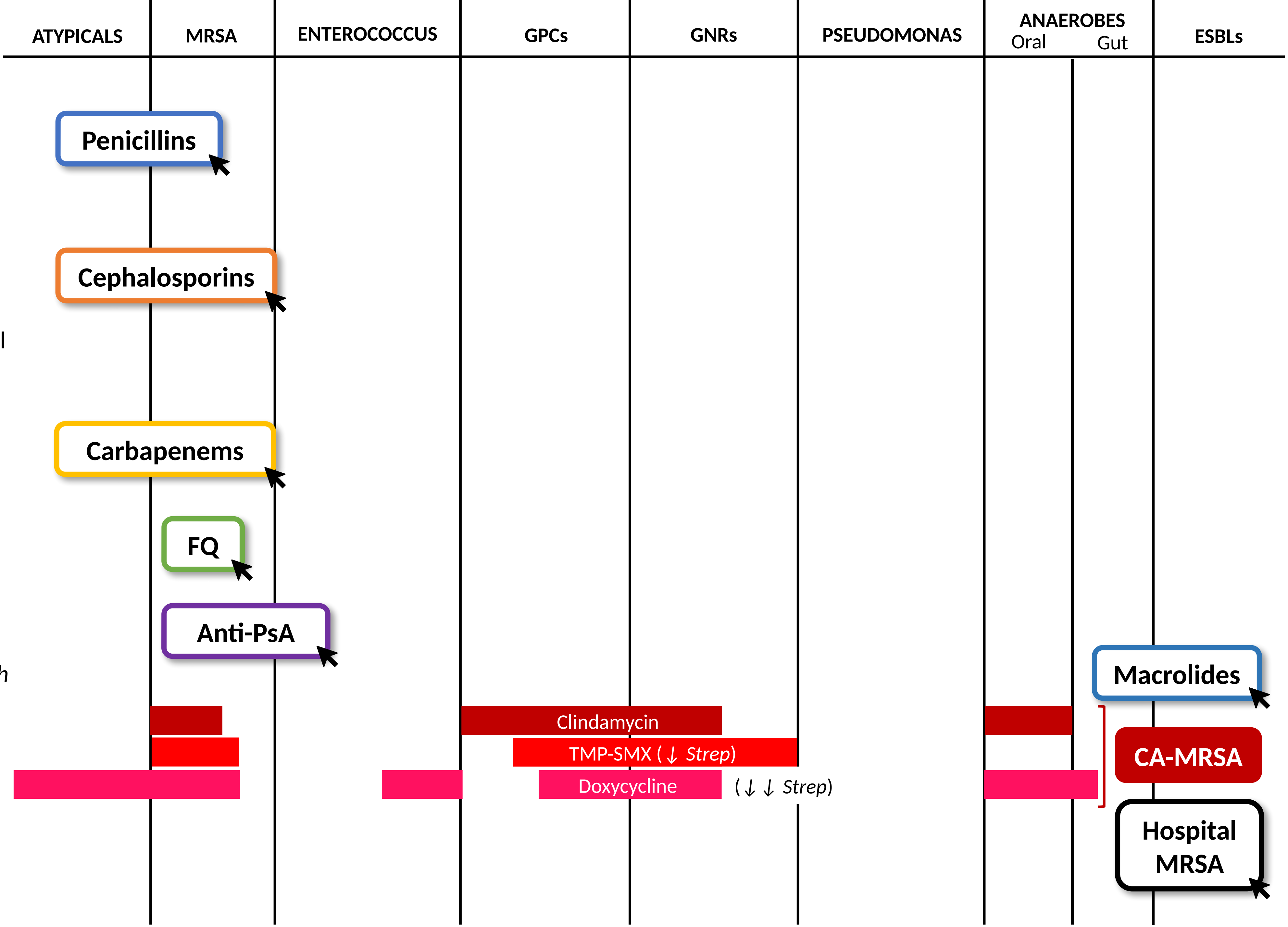
Strong oral anaerobe and GP coverage
No *Enterococcus*, gut anaerobes
Inhibits toxin production

TMP-SMX

Broad coverage of GNs and *Staph*
Variable *Strep* coverage

Doxycycline

Good atypical, oral anaerobic, and *Staph* coverage
Minimal *Strep* coverage



Basic Questions

Antibiogram

Full Board

Cases

General Rules:
Agents for nosocomial MRSA infections have pure GP coverage (including GP anaerobes)

Vancomycin

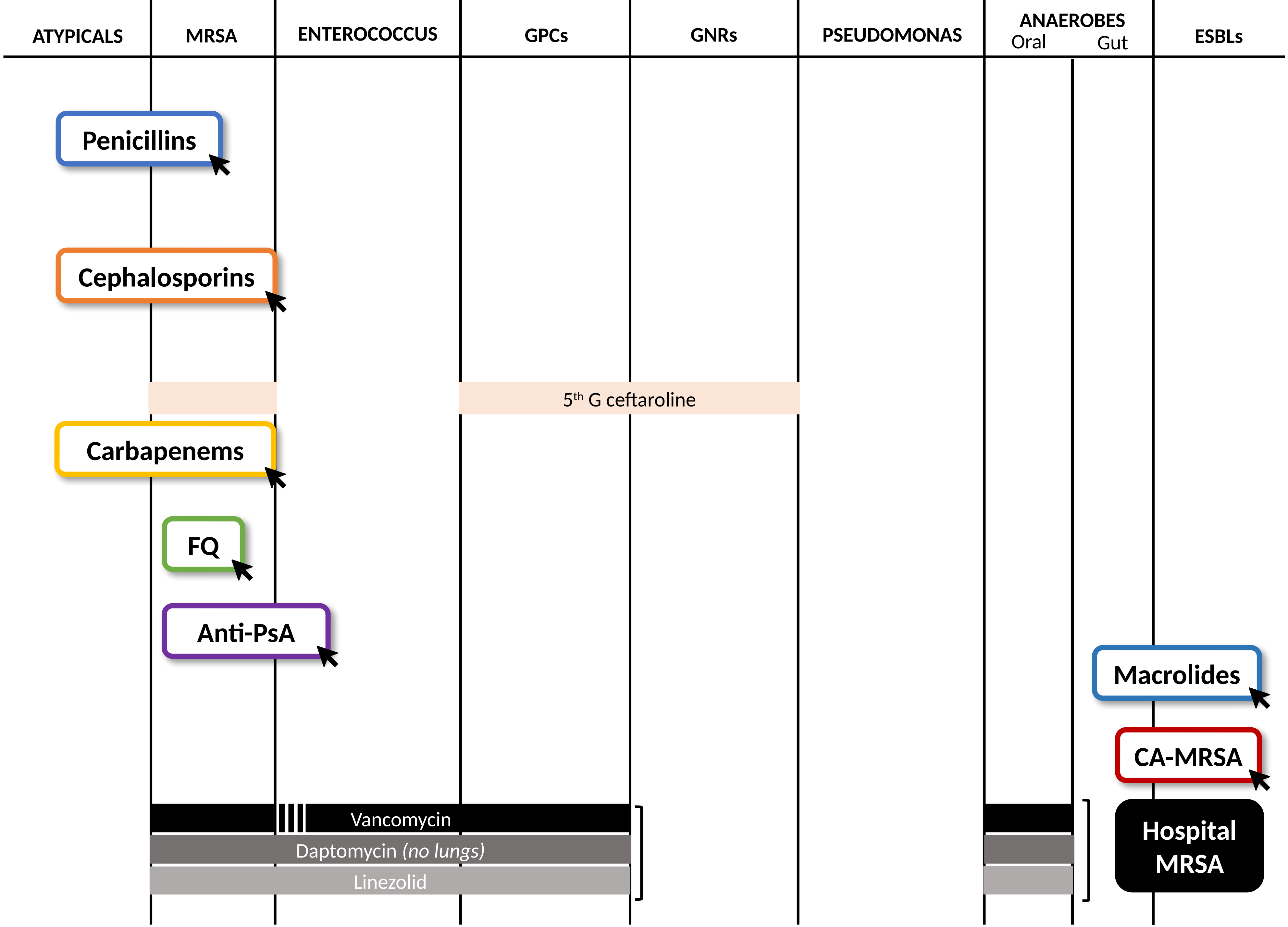
Only IV Vanc is used for MRSA
Some Enterococcus resistance (VRE)

Daptomycin

Cannot be used for MRSA pneumonia (deactivated by surfactant)

Linezolid

Treatment of choice for VRE, MRSA pneumonia



Basic Questions

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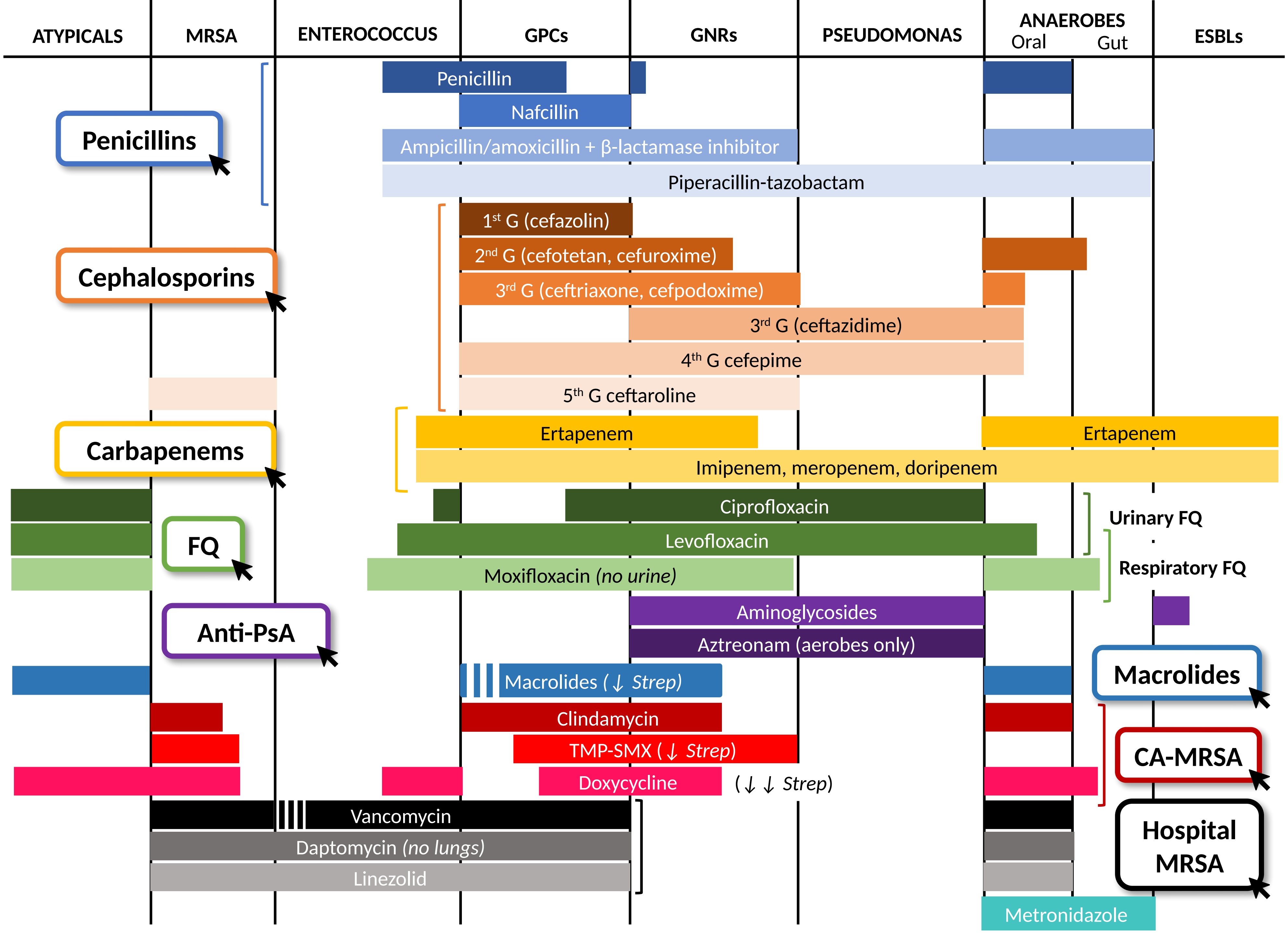

Case 1

31 yo with history of IVDU is admitted with sepsis from GP cocci bacteremia.

What antibiotics would you start?

IV Vancomycin. Good empiric coverage of GP organisms (including MRSA).

...Blood cultures speciate MSSA.

How would you adjust the antibiotics?

IV nafcillin. First line for MSSA infections.

Case 2

50 yo is admitted with an *E. coli* UTI. Sensitivities are pending.

What antibiotics would you start?

IV ceftriaxone. IV should be used here because the patient is septic.

What if they had a history of ESBL *E. coli?*

IV ertapenem. Carbapenems have the most reliable ESBL coverage.

Case 3

64 yo presents with RLL pneumonia. You want to empirically cover for *S. pneumoniae*, GNRs, and atypicals.

What antibiotics would you start?

Ceftriaxone (covers S. pneumo and GNRs) + **azithromycin** (covers atypicals). A respiratory FQ (e.g., levofloxacin) is also reasonable.

...MRSA nares swab is positive.

How will you adjust their antibiotics?

Add **IV Vancomycin** or **linezolid** (preferred in MRSA pneumonia). Daptomycin is inactivated by lung surfactant.

...You start IV vancomycin and they become red, hot, and flushed.

How will you adjust their meds?

This is VIR (vanc infusion reaction). Give an antihistamine and slow infusion rate.

Case 4

44 yo presents to clinic with purulent cellulitis, concerning for a MRSA infection. They have no signs of systemic infection.

What antibiotics would you start?

TMP-SMX, **clindamycin**, or **doxycycline.** Local sensitivities should dictate first line therapy.

Case 5

58 yo with an indwelling catheter is admitted for sepsis from a Pseudomonas UTI.

What antibiotics would you start?

If prior cultures are available, this should guide antibiotic therapy. Otherwise, empiric treatment with **urinary FQ (cipro, levo)** or **ceftazidime/ cefepime.** Pip-tazo and meropenem will cover PsA but are generally too broad for empiric therapy.

2024-2025 Context - Antibiotic Spectrum

- **Core spectrum/MOA teaching is timeless and was verified current (deck updated Mar 2024; expert-reviewed by ID).**
- **Newer/MDR Gram-negative agents to know (escalate with ID):**
  - Cefiderocol (siderophore cephalosporin) - covers carbapenem-resistant Enterobacterales, CRAB/Acinetobacter, Stenotrophomonas, PsA.
  - Newer beta-lactam/beta-lactamase-inhibitor combos: ceftazidime-avibactam, ceftolozane-tazobactam, meropenem-vaborbactam, imipenem-relebactam (KPC/AmpC/ESBL/PsA per agent). Aztreonam-avibactam for metallo-beta-lactamases.
- **Anti-MRSA additions: ceftaroline (the only beta-lactam with MRSA activity); long-acting lipoglycopeptides dalbavancin and oritavancin for ABSSSI (single/2-dose courses).**
- **C. difficile: fidaxomicin is now first-line over oral vancomycin (IDSA/SHEA 2021); deck already lists it.**
- **Unchanged high-yield rules: daptomycin inactivated by lung surfactant (never for MRSA pneumonia); linezolid often preferred for MRSA pneumonia; ertapenem has NO Pseudomonas coverage; carbapenems are most reliable for ESBL.**

Take-Home Points

For any antibiotic ask 4 questions: GPCs (incl. MRSA, Enterococcus)? GNRs (incl. Pseudomonas, ESBL)? Atypicals? Anaerobes (incl. B. fragilis/gut)?

Penicillins: no MRSA, no atypicals; GN coverage rises across classes (Pen -> antistaph -> amino+BLI -> pip-tazo). Pip-tazo = the one that adds Pseudomonas.

Cephalosporins: no Enterococcus, no atypicals; GN rises by generation. Ceftazidime/cefepime add PsA; ceftaroline (5th gen) covers MRSA; no oral cephalosporin covers PsA.

Carbapenems: broadest (GP/GN/anaerobe), most reliable ESBL; no MRSA/atypical, poor Enterococcus. Ertapenem = once-daily, NO Pseudomonas; mero/imi/dori add PsA.

FQs: cipro (urinary, GN/PsA, poor Strep); levo (urinary+respiratory, adds S. pneumo); moxi (respiratory, best anaerobe, weak PsA/renal).

MRSA: CA -> clindamycin / TMP-SMX / doxycycline; HA -> vancomycin (1st-line), daptomycin (NOT for pneumonia), linezolid (VRE, MRSA pneumonia).

Attribution & Changelog

Adapted from TeachIM "Antibiotics - Part 1: Spectrum" (teachim.org), updated March 2024.

- Authors: Brandon Fainstad MD; Yilin Zhang MD. Expert reviewer: Diana Zhong MD MHS.
- Source: teachim.org/teaching_material/antibiotics-part-1-spectrum/ | Deck: Antibiotic-spectrum-final-3.5.24.pptx

Corrections (this optimization, 2026-06-10):

- "cefofetan" -> "cefotetan" (correct spelling of the 2nd-gen cephamycin) on slides 5, 12, 14. All click-to-reveal animations and graphics preserved.

Added: 2024-25 newer-agent context slide, take-home slide, this attribution slide.

- Companions: Learner-Worksheet.pdf (fill-in antibiogram) and Antibiogram-4.22.pdf - reviewed, no changes. Newer context source-cited; clinician/ID-review before teaching.